

Introduction to Data Science
Final Project Report

Traffic Volume Prediction Using Atmospheric Reanalysis Data

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1 Introduction

In this project, we aim to evaluate the feasibility of using atmospheric reanalysis (estimated historical global climate data) to predict traffic volume, in contrast to local measurement and observation of weather conditions.

We'll first gather traffic volume data coupled with local weather measurements, and gather atmospheric reanalysis data at place of measurement. Then, we'll build regression models that predict traffic volume from weather data from each of the datasets, and other prediction variables such as hour of day, day of the week, etc. Finally, we'll compare and contrast the goodnesses of fit obtained from each regression method and weather dataset used.

2 Data Collection

The datasets used in this project include:

- Metro Interstate Traffic Volume
(<https://archive.ics.uci.edu/dataset/492/metro+interstate+traffic+volume>)
- ERA5 hourly data on single levels from 1940 to present
(<https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels>)

First, the Metro Interstate Traffic Volume data (which will henceforth be referred to as “**ATR**”, short for Automatic Traffic Recorder) details the traffic volume data from 2012 to 2018 at a spot in interstate highway I-94, Minnesota, United States. The dataset also contains local weather and holiday observations.

And second, the ERA5 hourly data (“**ERA5**”) is a meteorological reanalysis that estimates various climate parameters from historical data with a resolution of 0.25° from 1940 onwards. The data was requested year-by-year as a GRIB file, that was later converted to CSV using a custom-written python script. Finally, the coordinate was obtained by utilizing the Roadway Project Mapping Application (<https://www.dot.state.mn.us/roadway/data/rpma.html>) to exactly locate the whereabouts of MN DoT ATR station 301 that provides the ATR dataset, which is in route I-94, reference post 239+00.922.

3 Data Exploration

Now that the two datasets have been obtained, let's look at the relationships between variables to see what insights we could gather from. This section will be divided into 4 parts, as follows:

1. Variables tables
2. Data distributions and relationships
3. Data correlations with traffic volume
4. Insights

3.1 Variables tables

Table 1: Variables table of ATR dataset

(From <https://archive.ics.uci.edu/dataset/492/metro+interstate+traffic+volume>)

Variable	Type	Units	Description
holiday	String		US National holidays and regional holidays
temp	Float	Kelvin	Average temperature in the hour
rain_1h	Float	mm	Rainfall in the hour
snow_1h	Float	mm	Snowfall in the hour
clouds_all	Integer	%	Percentage of cloud cover
weather_main	String		Short description of current weather
weather_description	String		Detailed description of current weather
date_time	Datetime		Hour of data collected in local CST time
traffic_volume	Integer		Reported hourly westbound traffic volume

Table 2: Variables information of ERA5 dataset

(From <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels>)

Variable	Name	Type	Units
t2m	2m temperature	Float	Kelvin
tcc	Total cloud cover	Float	
lcc	Low cloud cover	Float	
mcc	Medium cloud cover	Float	
hcc	High cloud cover	Float	
valid_time	Valid datetime	Datetime	
pptype	Precipitation type	Integer	
tp	Total precipitation	Float	m
sf	Snowfall	Float	m
cp	Convective precipitation	Float	m
lsp	Large-scale precipitation	Float	m

Variable	Short Description
t2m	Temperature at 2m above the surface
tcc	Proportion of cloud clover in total (0-1)
lcc	Proportion of cloud clover at lower levels (0-1)
mcc	Proportion of cloud clover at medium levels (0-1)
hcc	Proportion of cloud clover at higher levels (0-1)
valid_time	Hour of data in UTC time
pptype	Precipitation type at surface, assigned with integers
tp	Accumulated rain and snow in the hour
sf	Accumulated snow in the hour
cp	Accumulated convective precipitation in the hour
lsp	Accumulated large-scale precipitation in the hour

3.2 Data distributions and relationships

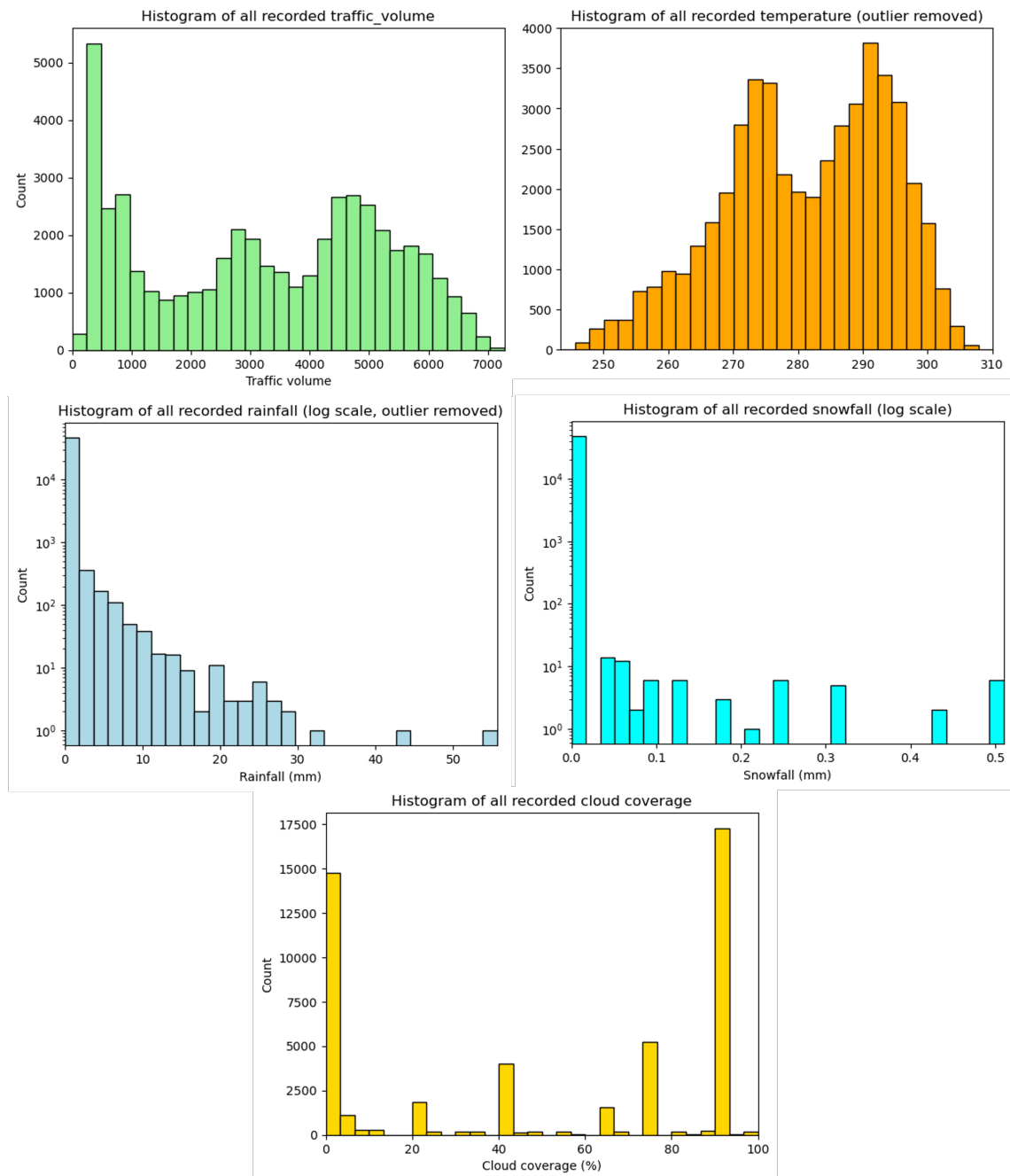


Figure 1: Selected distribution of data in ATR dataset

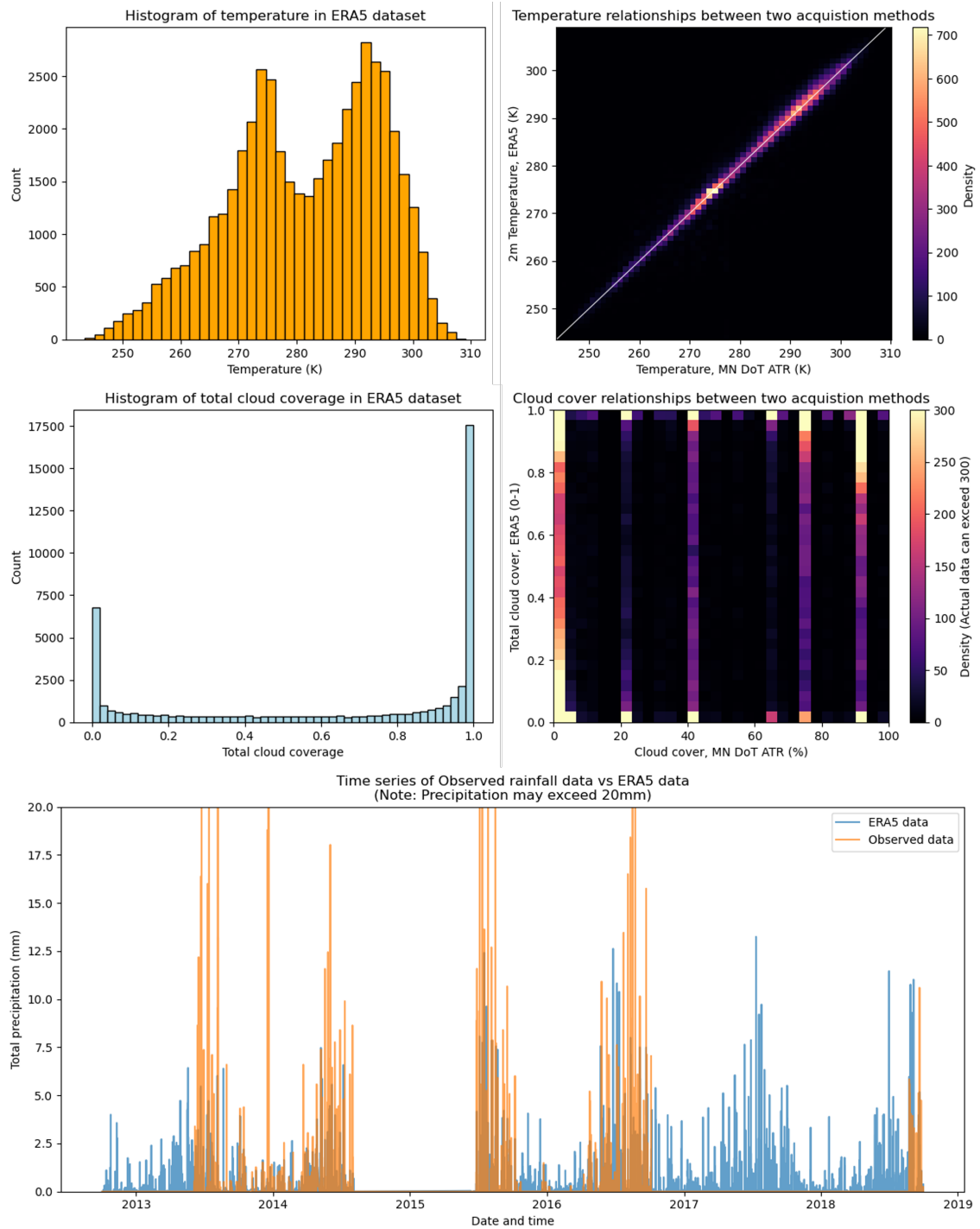


Figure 2: Selected distribution of data in ERA5 dataset, and few relations between the two datasets

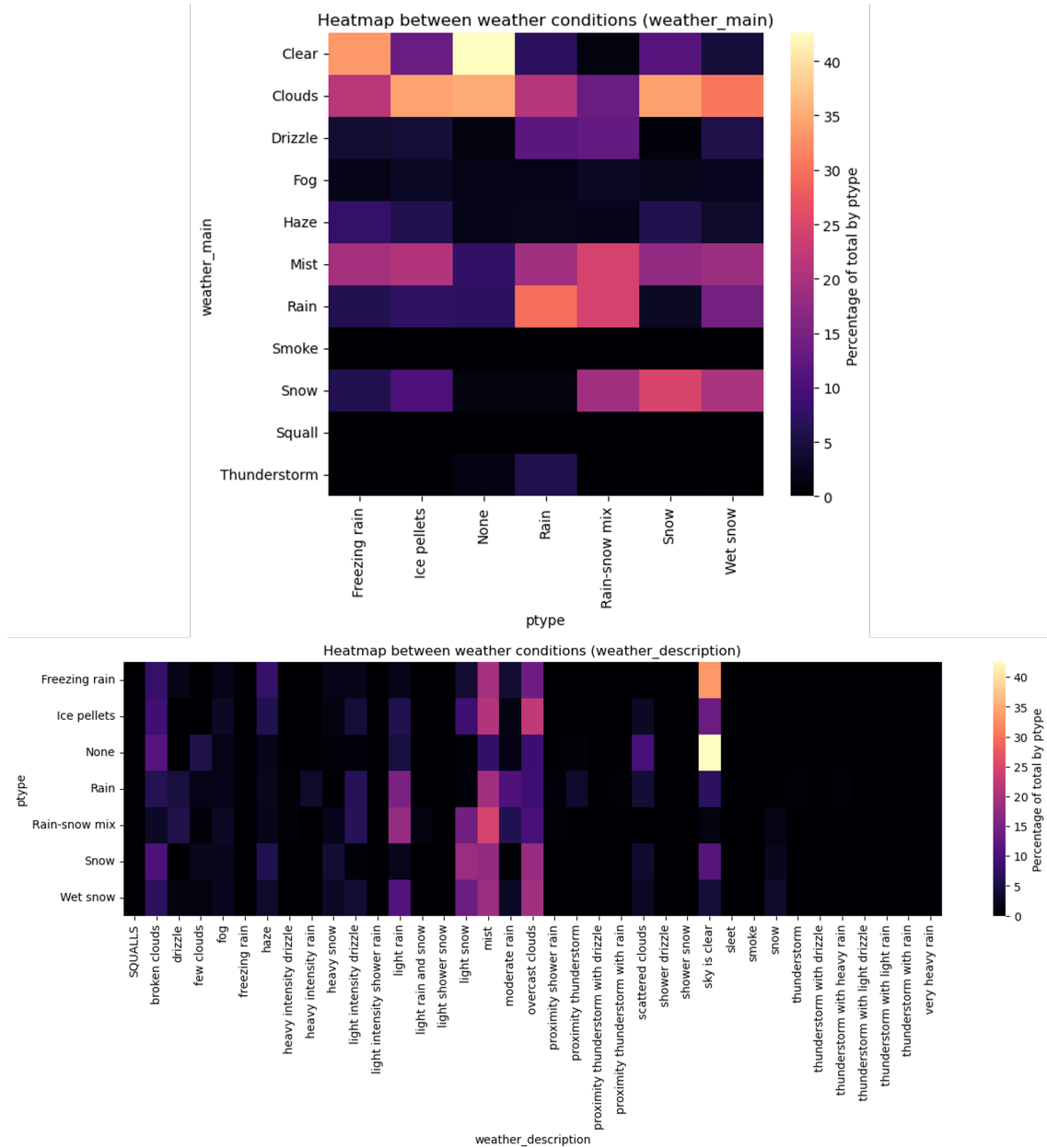


Figure 3: Heatmaps between different weather descriptions (ATR, 11 if main, 38 if detailed) and precipitation types (ERA5, 7 in total)

3.3 Data correlations with traffic volume

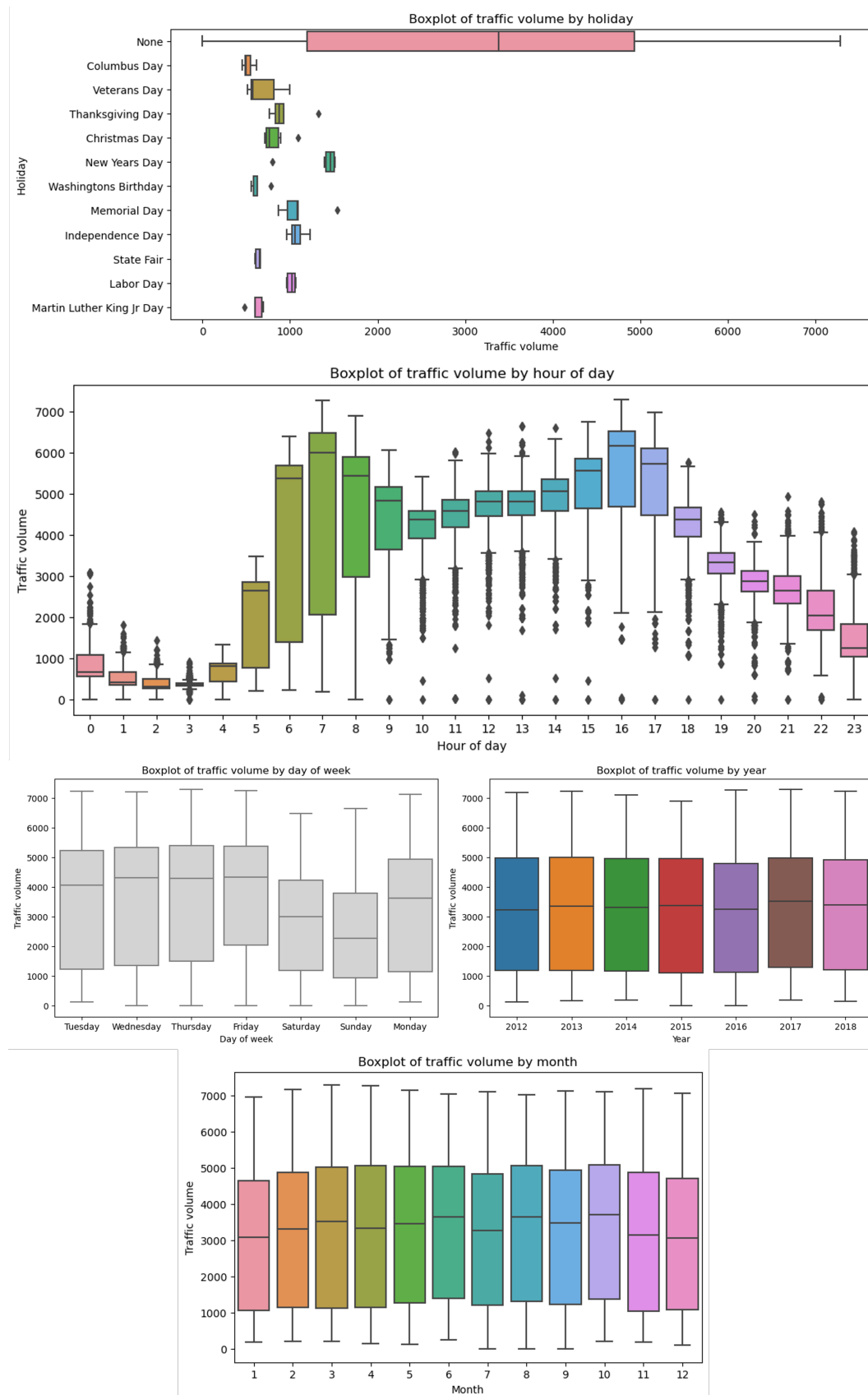


Figure 4: Traffic volume distribution by non-weather variables from ATR

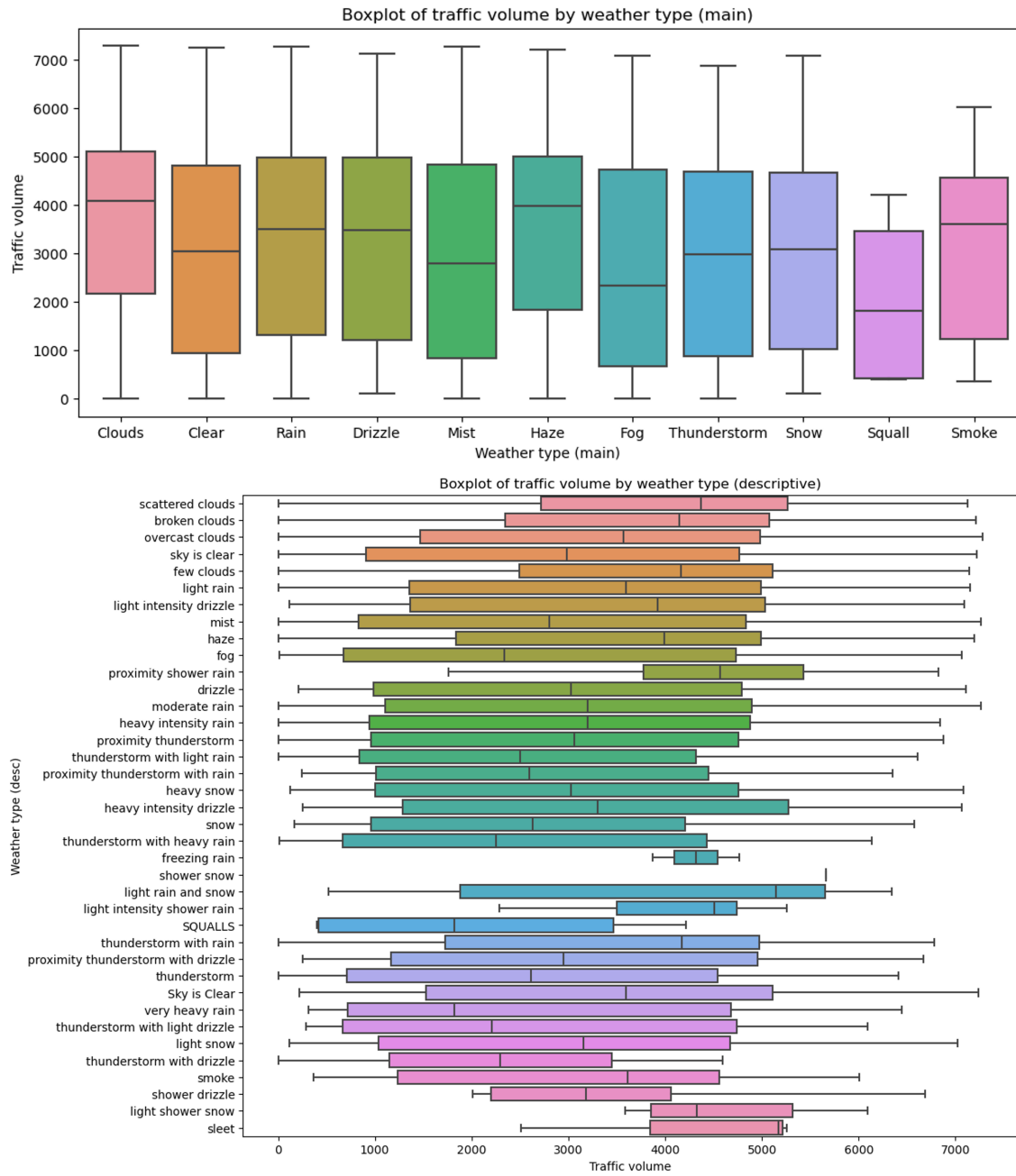


Figure 5: Traffic volume distribution by weather descriptions from ATR

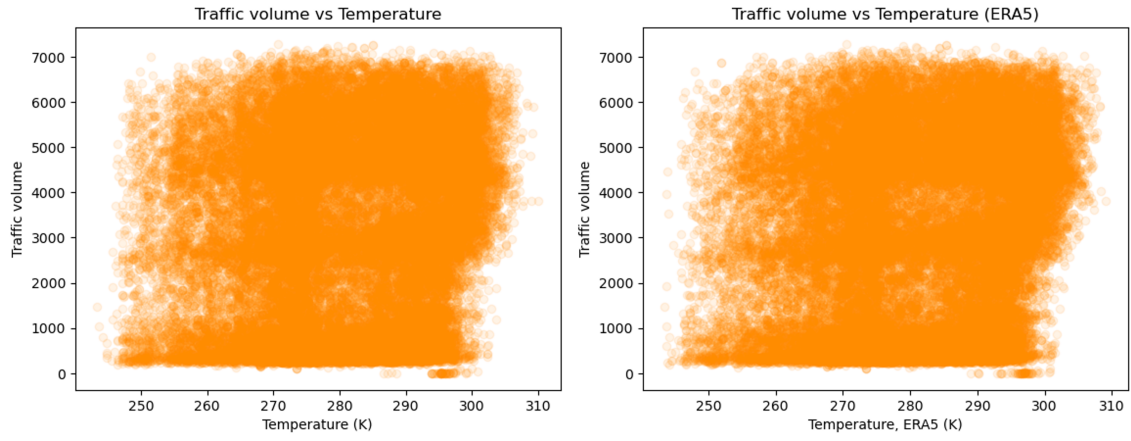


Figure 6: Traffic volume distribution by temperature from both datasets

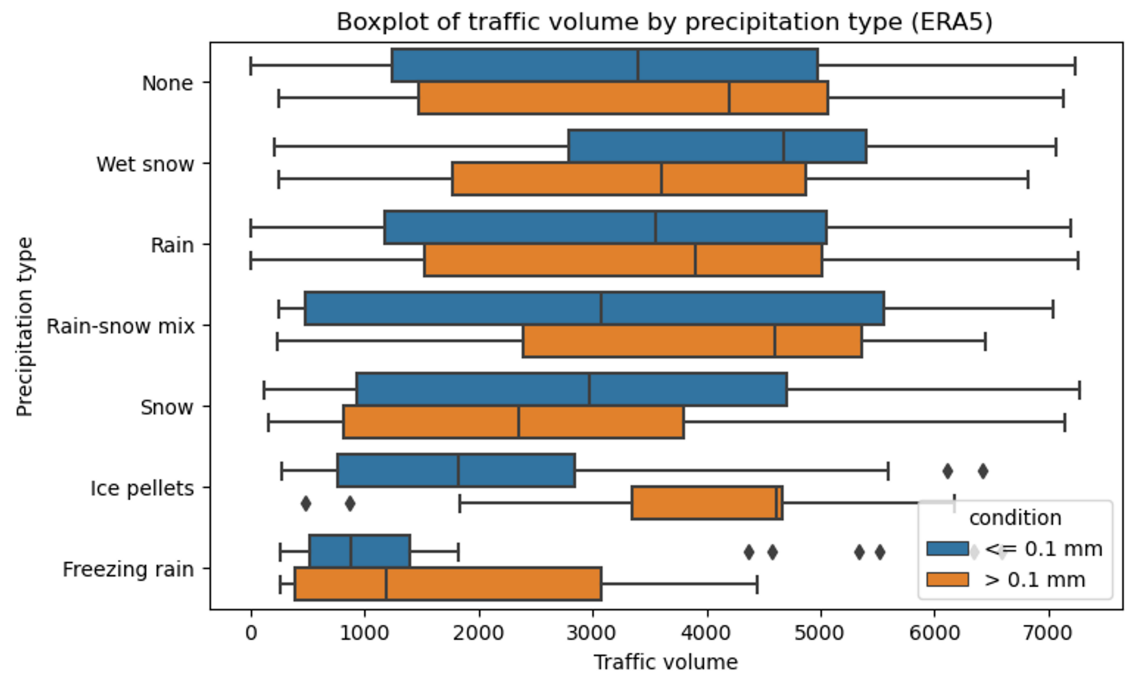
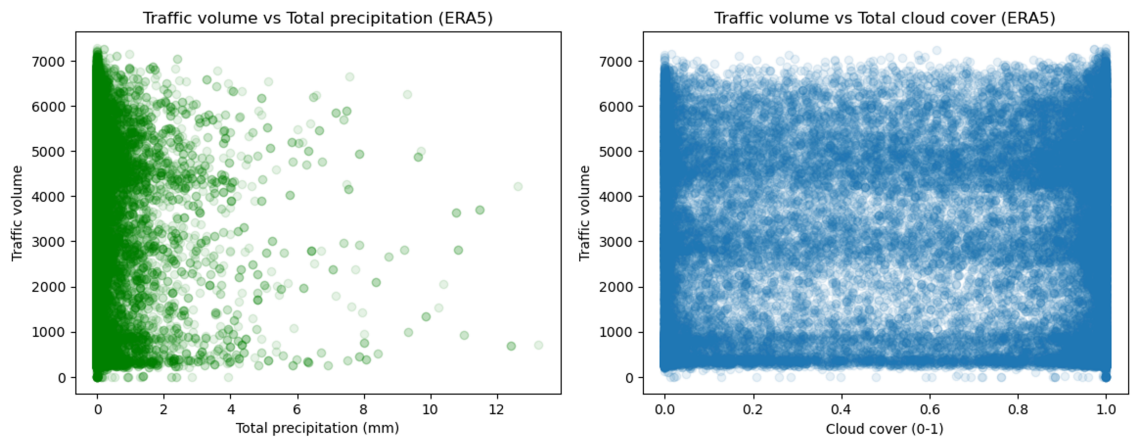


Figure 7: Traffic volume distribution by variables from ERA5

3.4 Insights

From the data exploration process summarized as graphs above, one could see that

- The recorded traffic volume has a range of approximately 0 - 7000.
- The temperatures obtained from two datasets are almost identical. This suggested the ERA5 data obtained was indeed in the same location at the same timeframe.
- The precipitation data from ATR is likely unusable. Apart from the missing data in 2015, the data suggested that it did not rain nor snow once there during 2017, which is unlikely.
- Weather conditions from ATR looks to be related to precipitation types from ERA5. For instance, most of the “None” precipitation types in ERA5 correspond to “Clear” and “Sky is Clear” main and detailed weather descriptions in ATR.
- Holiday, hour of day, and weekends are non-weather variables that significantly impact traffic volume.
- The traffic volume stayed relatively constant for all months of the year and years.
- Weather types and precipitation types for both ATR and ERA5 significantly affect traffic volume. For ERA5, the effect is more pronounced if we were to separate the precipitation types with a threshold of 0.1 mm.
- Temperature, cloud cover, and total precipitation has a little-to-no correlation with traffic volume.

4 Model Selection and Evaluation

4.1 Regression methods

Now that we have explored the data, the prediction variables will be

- Numerical: Temperature
- Categorical: Hour of day (24), Holidays (11), Days of the week (7), and Weather type (see *Weather data approaches*)

All categorical variables are encoded via one hot encoding.

Next, as for the regression methods, we’ll compare the following 5 models:

1. Linear regression
2. Decision tree
3. Random forest
4. Bagging
5. XGBoost

Hyperparameters are not adjusted for all models, so overfitting is expected, especially in decision tree models. This is sufficient as we only wish to compare the performance between weather data acquisition methods used in a model.

Finally, the data is split into training data and test data with a ratio of 7:3, where the test data encompasses the last 30% of the data by date. This is to simulate the prediction of future traffic, and not interpolate traffic volume data.

4.2 Weather data approaches

For each regression model, we'll evaluate the goodnesses of fit for each of the following weather data included:

- No weather considerations [0]
- Main weather description (ATR) [11]
- Detailed weather description (ATR) [38]
- Precipitation type, separated by 0.1 mm threshold (ERA5) [14]

Why 0.1 mm as the threshold? The answer is that while it's completely arbitrary, it's roughly an order of magnitude that sufficiently separates between "heavy precipitation" and "light precipitation" without splitting the data too unevenly. And from the data exploration results, separating precipitation type by total precipitation does create distinct traffic volume results.

Moreover, apart from the usual train and test data RMSE, we'll also evaluate "Bad weather test RMSE," defined as the RMSE from the data at only bad weather, or "Clear" and "Clouds" main weather description (ATR) excluded. This is because we're interested in predictions during bad weather specifically, and is the reason why weather conditions are included as prediction variables in the first place.

5 Results

The following are the graphs which summarizes the goodnesses of fit for each regression method and weather data approach:

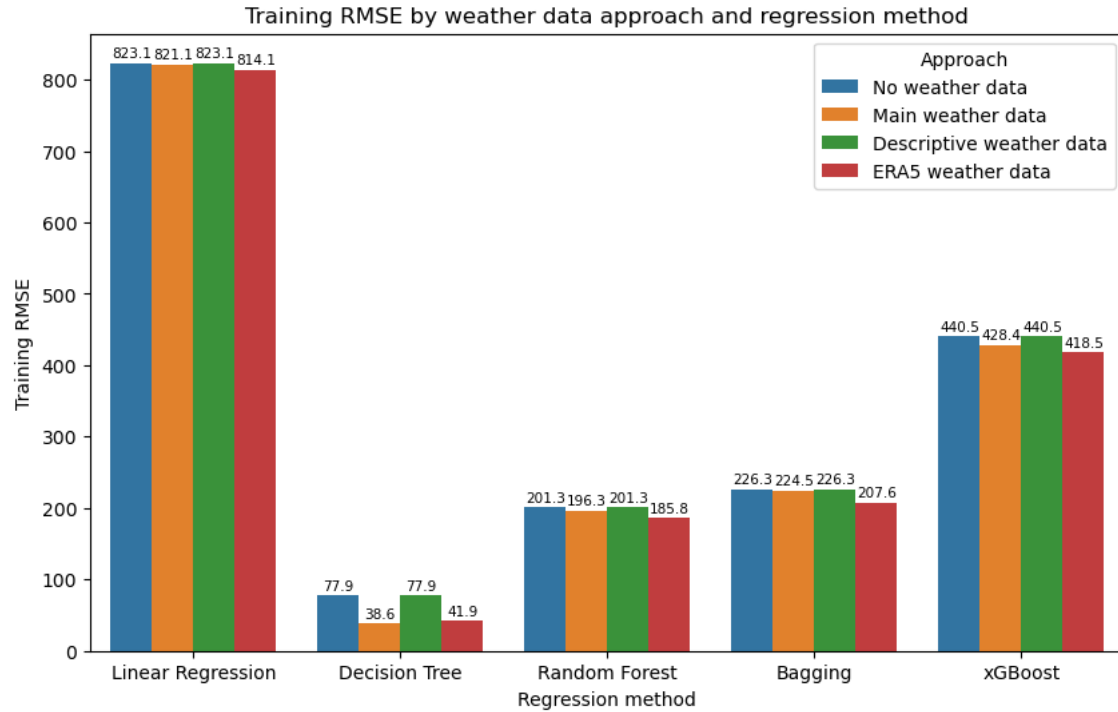


Figure 8: Training RMSE, separated by regression method and weather data approach

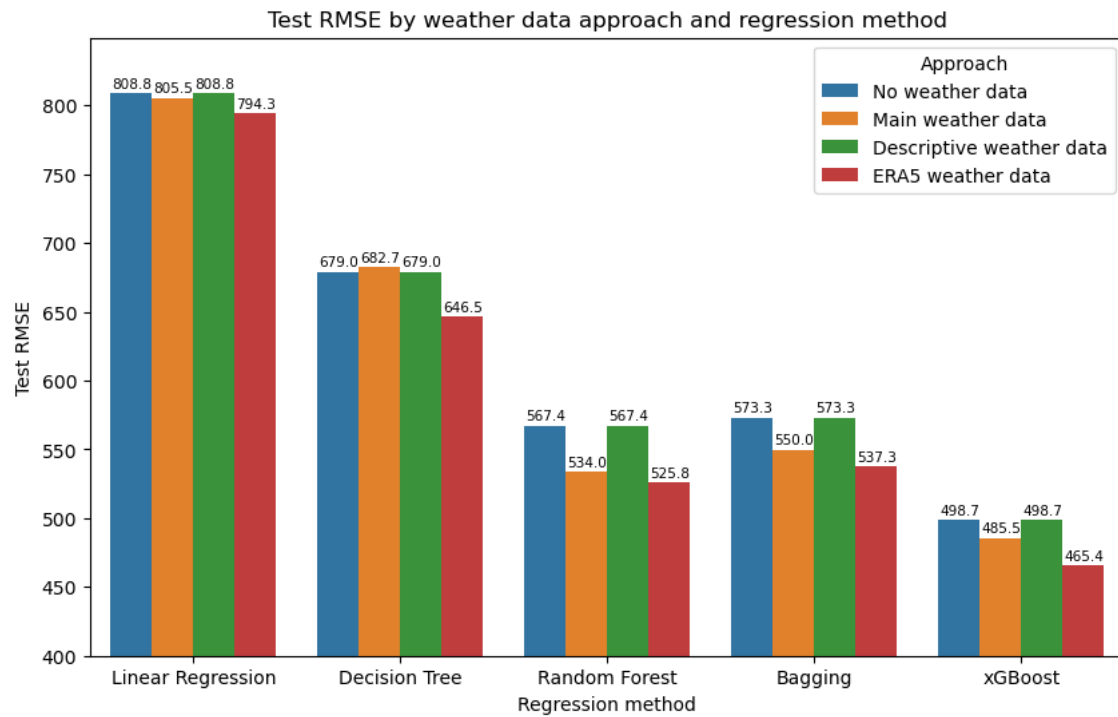


Figure 9: Test RMSE, separated by regression method and weather data approach

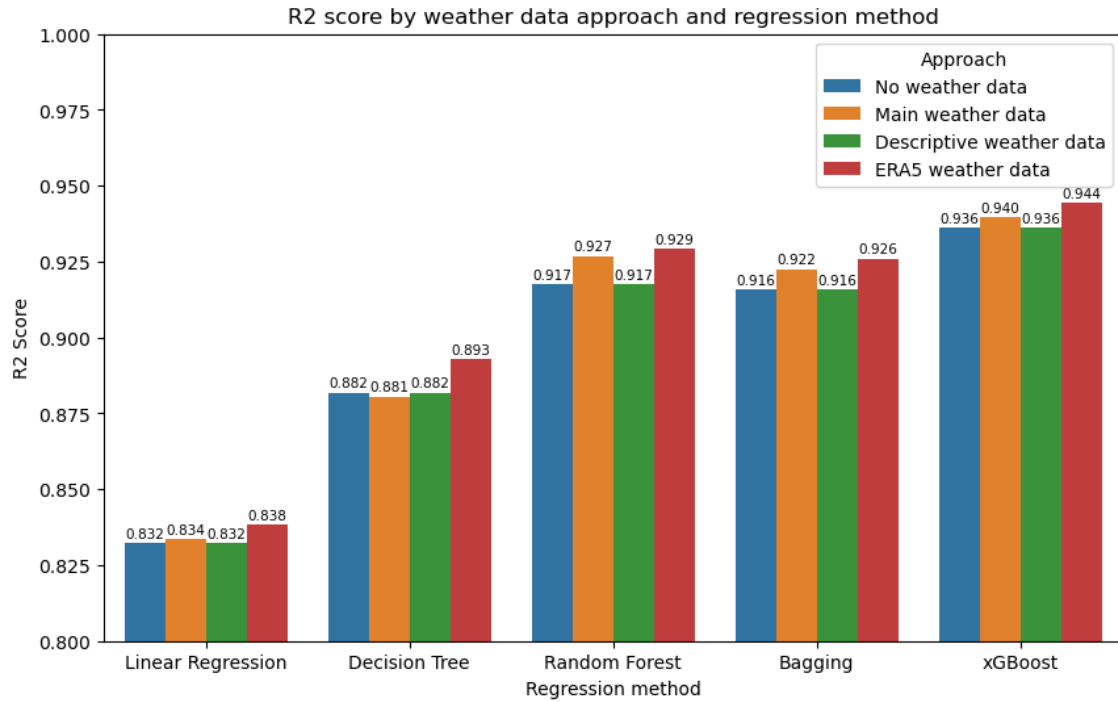


Figure 10: Test R^2 score, separated by regression method and weather data approach

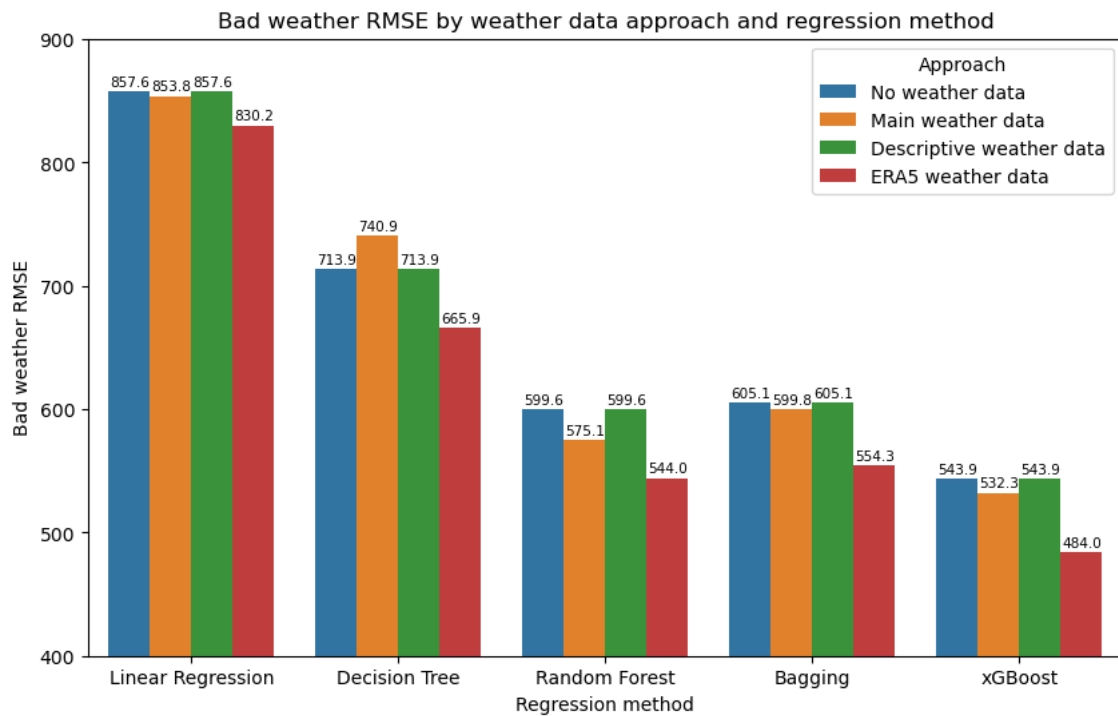


Figure 11: Test RMSE during bad weather, separated by regression method and weather data approach

6 Discussion

From the results, one could see that

- By test RMSE performance, the best models ordered by descending RMSE are: XGBoost, Random forest, Bagging, Decision tree, and Linear regression
- Decision tree, random forest, and bagging regression methods clearly overfit the training data. This is due to not having set the hyperparameters.
- Some data, such as convective precipitation (which implies heavier rain) large-scale precipitation data (which implies lighter rain) from ERA5 data were not utilized. Meaning that the model could be improved further if these variables (and even values not taken from ERA5 request calls) were taken into account.
- Unexpectedly, local weather measurements data did not improve the RMSE of the model at all. In fact, detailed weather descriptions made the model slightly worse.
- Models that use ERA5 data outperformed local weather measurements in all regression methods by up to 6% in test RMSE overall, and 11% in test RMSE during bad weather.

7 Conclusion

In conclusion,

- This project aims to predict traffic volume based on atmospheric reanalysis (ERA5) data and compare the performance with local weather measurements.
- Weather type has a significant impact on traffic volume, with models accounting for atmospheric reanalysis data outperforming predictions that account local weather observations by up to 6% in RMSE overall, and up to 11% in test RMSE during bad weather.
- Things not accounted for in the project include regression models requiring hyperparameter tuning to not overfit the training data, and some column data not being used in the prediction model.
- This showed that atmospheric reanalysis is a feasible substitute for local weather measurements when building prediction models that require local weather data.